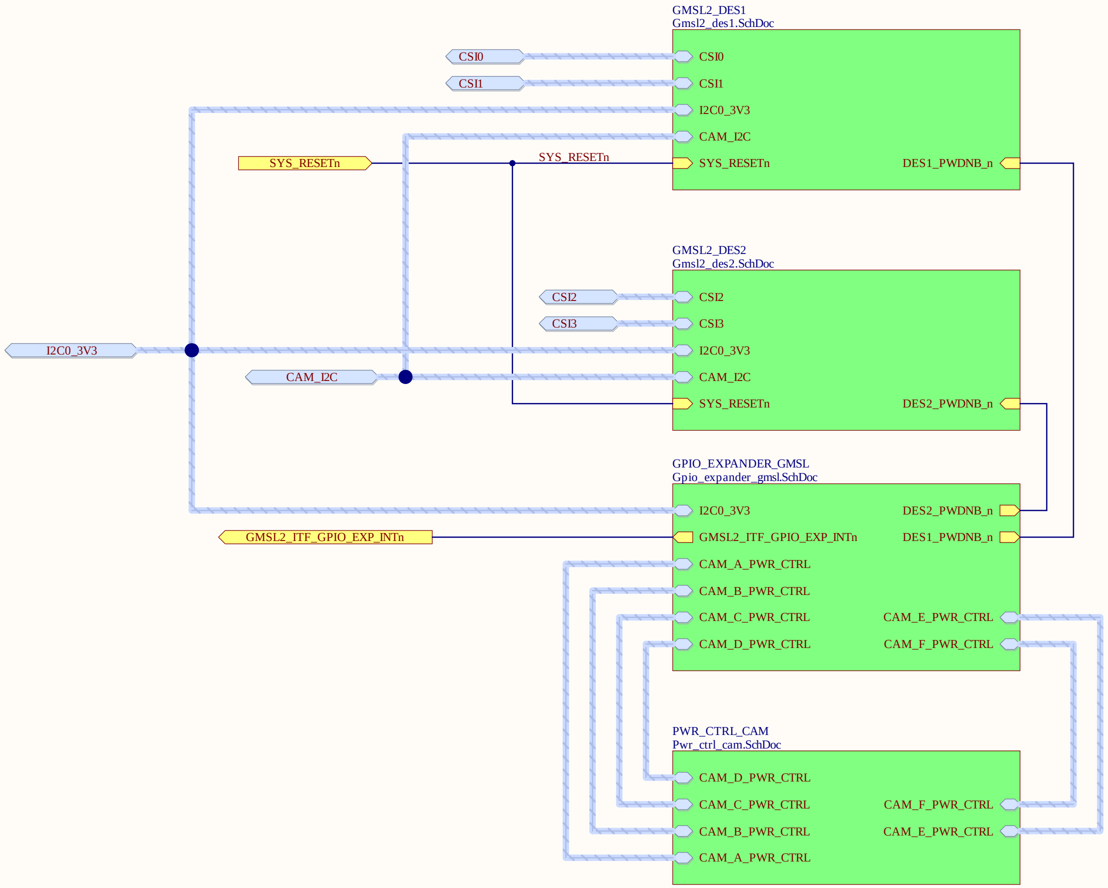

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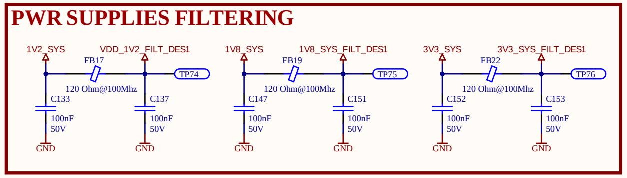
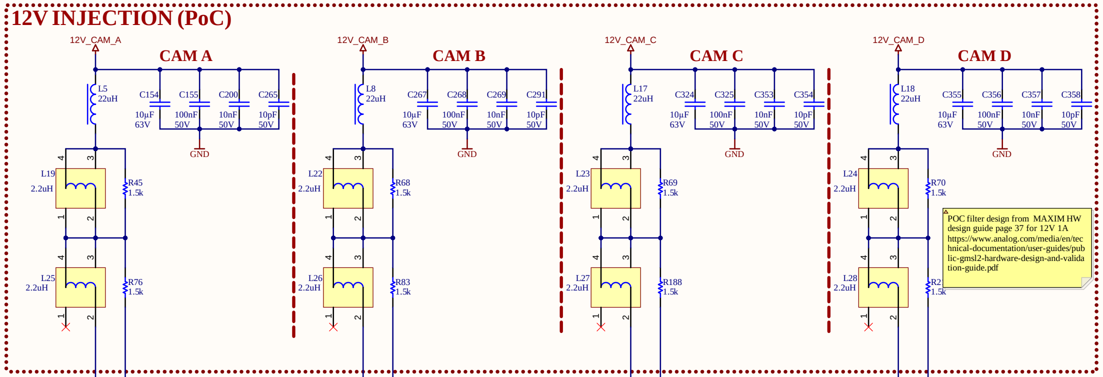




POC filter design from MAXIM HW design guide page 37 for 12V 1A https://www.analog.com/media/en/technical-documentation/user-guides/public-gmsl2-hardware-design-and-validation-guide.pdf

Input Data rate calculation (SIO Side) :
Camera is ZED X One GS Wide. Output format is RAW10 (from camera datasheet) which means 10bits/pixel.
Preferred capture mode is 960x600@60fps (from Presien Specification)
It leads to 960x600x10x60 = 345.6Mbps per camera video + Overhead Protocol

Layout note : CSI Diff Pair should be 100R Diff + All CSI Lane whatever the bus should be lenght matched all together. See picture below

Table 10-4. MIPI CSI D-PHY Interface Signal Routing Requirements

Parameter	Requirement	Units	Notes
Max Data Rate (per data lane) for High-Speed mode Max Frequency (for Low Power mode)	2.5 10	Gbps MHz	
Number of loads	1	load	
Reference plane	GND		
Trace impedance: Diff pair / SE	90-100 / 45-50	Ω	±10%
Via proximity (signal to reference)	< 0.65 (3.8)	mm (ps)	
Intra-pair trace spacing	0.15mm	mm	Can be adjusted to meet Differential Impedance.
Trace spacing: Microstrip / Stripline	2x / 2x	dielectric	
Max PCB breakout delay	48	ps	
Max Insertion loss 1 Gbps 1.5 Gbps 2.5 Gbps	 3.00 2.90 1.92	 dB	
Max trace delay / length 1 Gbps (Stripline/Microstrip) 1.5 Gbps 2.5 Gbps	 2526 (421) / 2487 (421) 1913 (319) / 1885 (319) 900 (150) / 886 (150)	 ps (mm)	
Max intra-pair skew	1	ps	
Max trace delay skew between DQ and CLK 1 / 1.5 / 2.5 Gbps	40 / 26.7 / 16	ps	**DQ** includes all the data lines associated with a single clock. This may be 2 differential data lanes for a x2 interface, or 4 differential data lanes for a x4 interface.
Keep critical traces away from other signal traces or unrelated power traces/areas or power supply components			

Layout : Remove "coper ring" in internal layer to avoid capacitive effects

Layout note : Place link capacitor as close as possible to its corresponding SIO Pin

All serial cap are 100nF 0402 50V
Serial resistor are 49.9R 0402

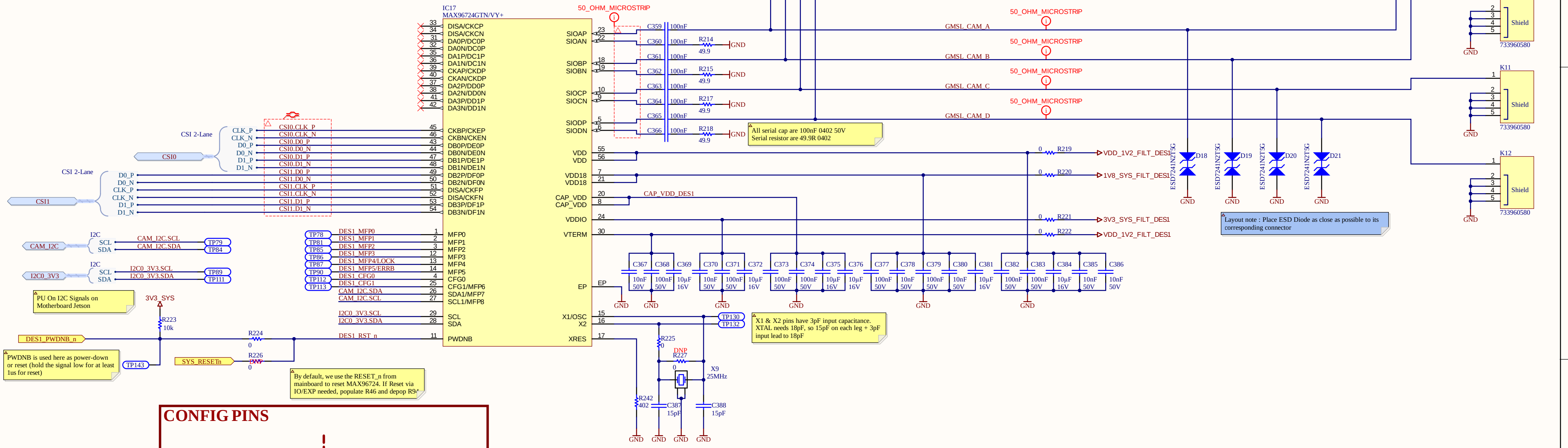
Layout note : Place ESD Diode as close as possible to its corresponding connector

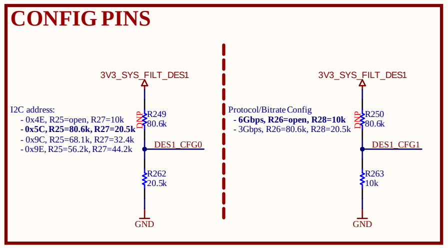
PU On I2C Signals on Motherboard Jetson

PWDNB is used here as power-down or reset (hold the signal low for at least 1us for reset)

By default, we use the RESET_n from mainboard to reset MAX96724. If Reset via IO/EXP needed, populate R46 and depop R9/

X1 & X2 pins have 3pF input capacitance.
XTAL needs 18pF, so 15pF on each leg + 3pF input lead to 18pF





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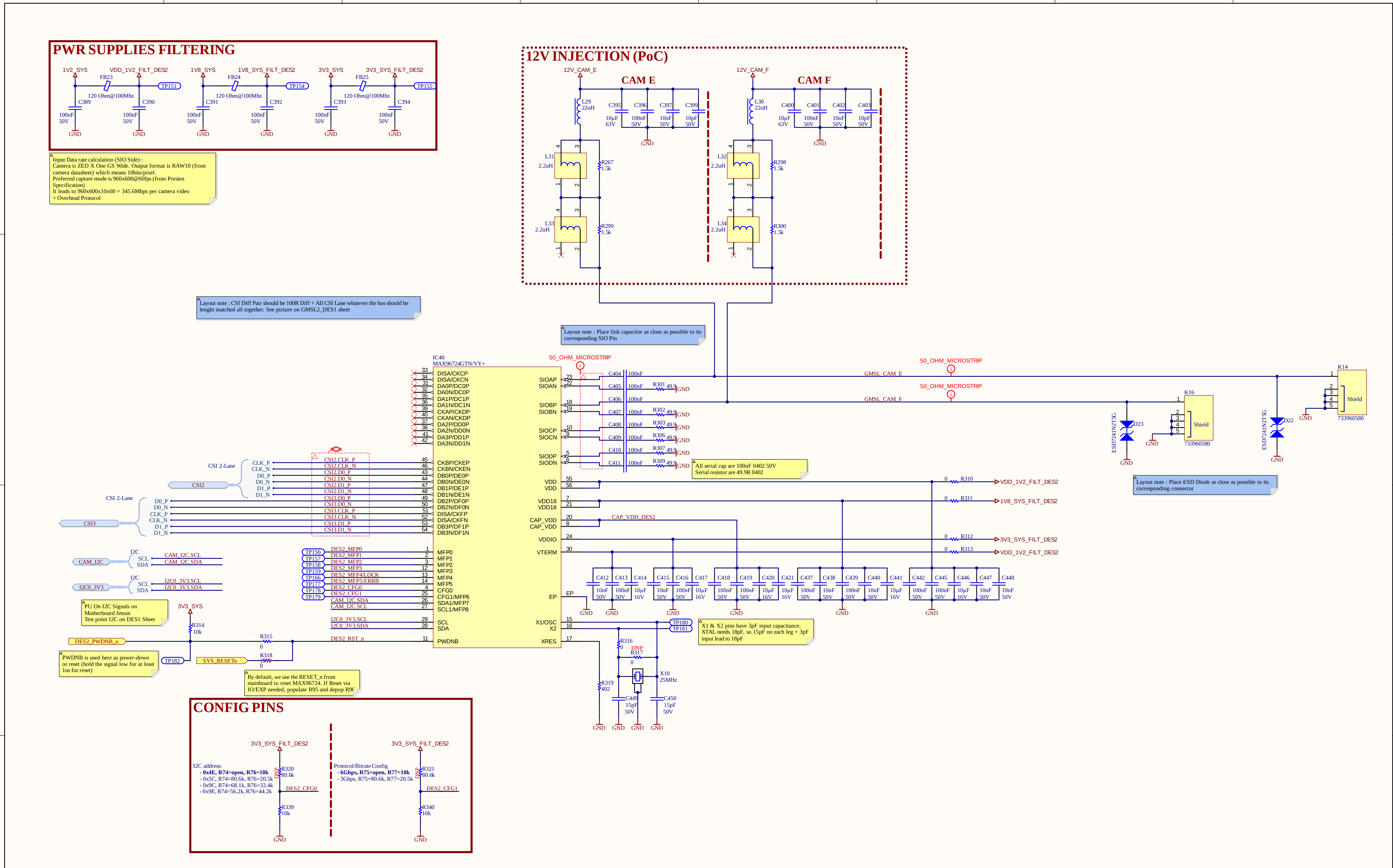

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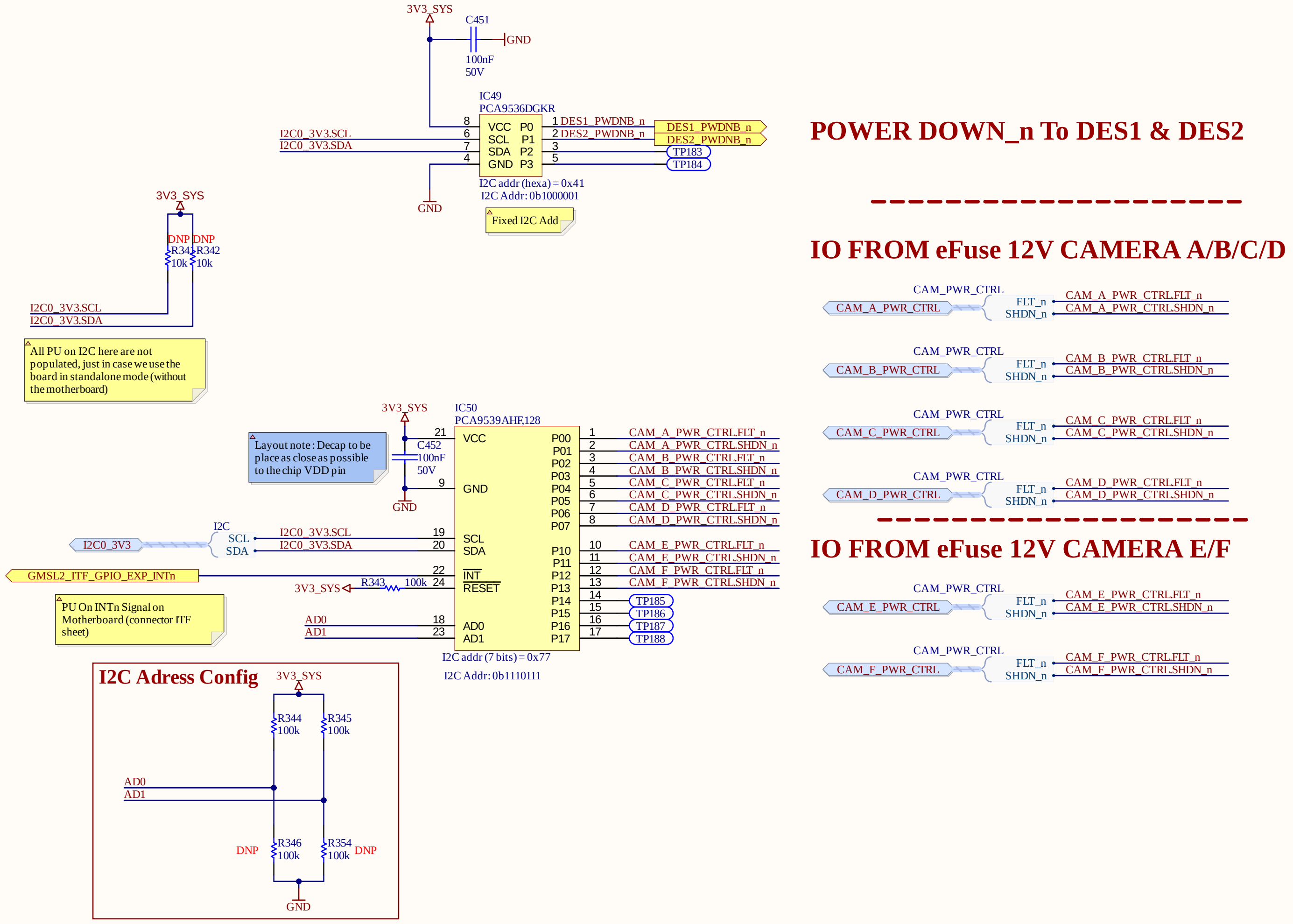

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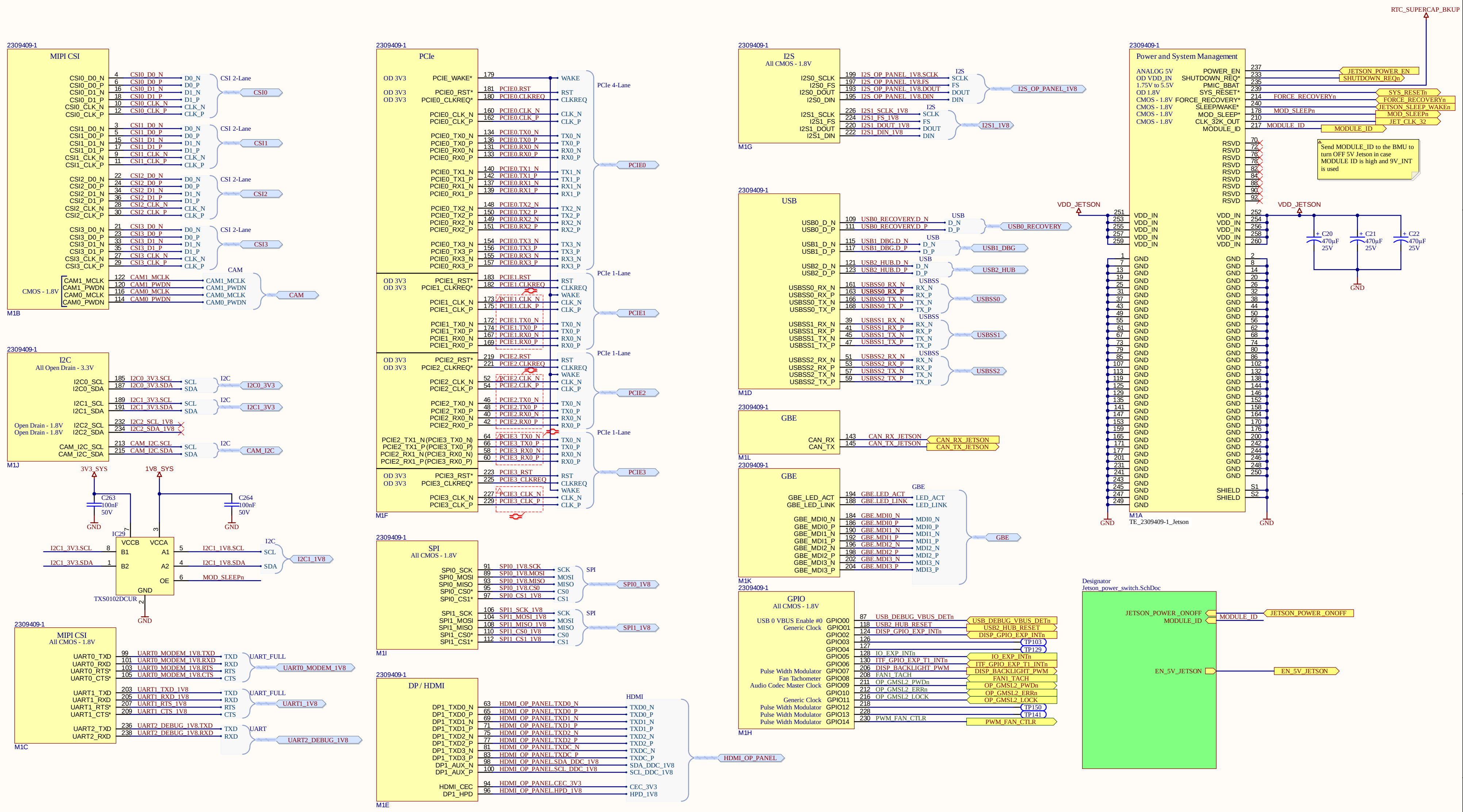

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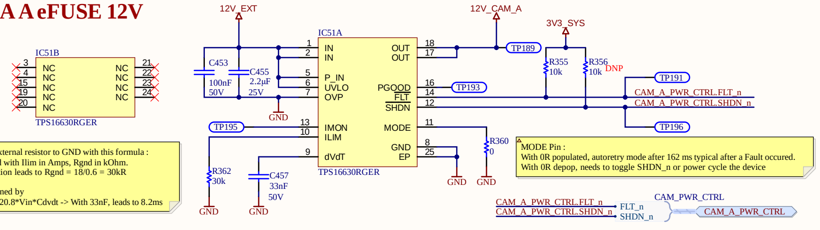
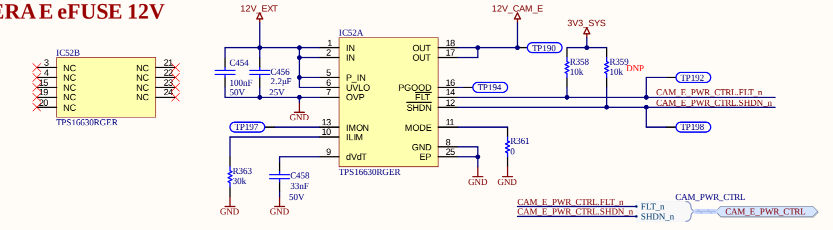
CAMERA A eFUSE 12V



CAMERA E eFUSE 12V

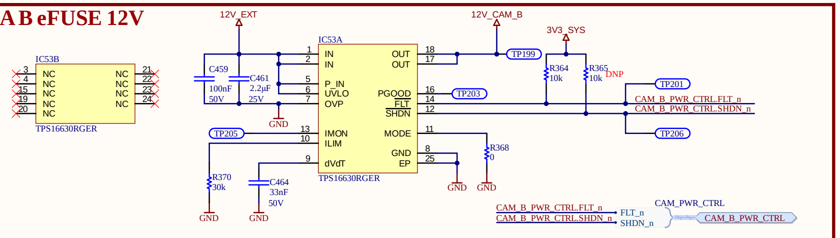


Power Consumption of Camera ZED X ONE GS WIDE is :
- Min : 0.7W <==> 59mA @12V
- Max : 0.94W <==> 79mA @12V

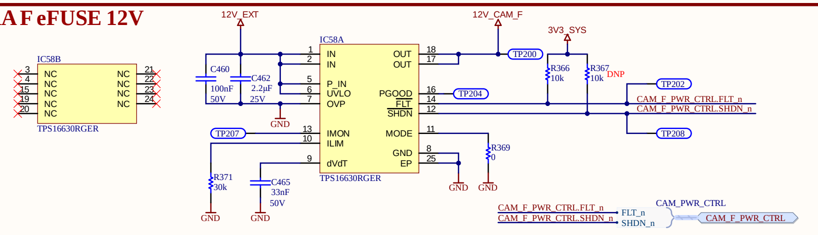
For reference :
https://www.stereolabs.com/docs/get-started-with-zed-link/power_requirements

For later design change, if we need to reduce the price, use the following eFuse : TPS1641x.

CAMERA B eFUSE 12V



CAMERA F eFUSE 12V



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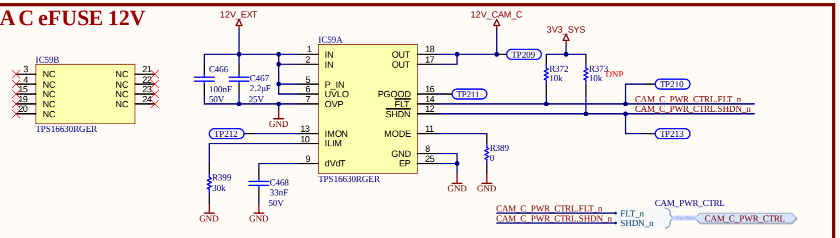
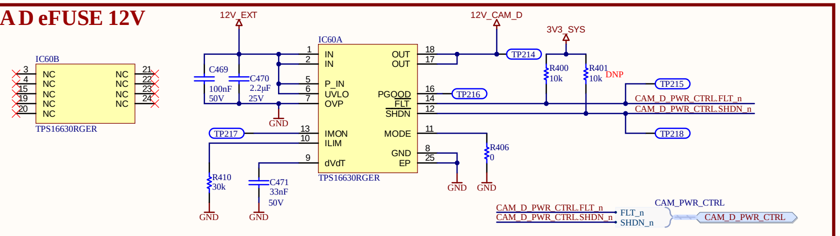


CAMERA D eFUSE 12V



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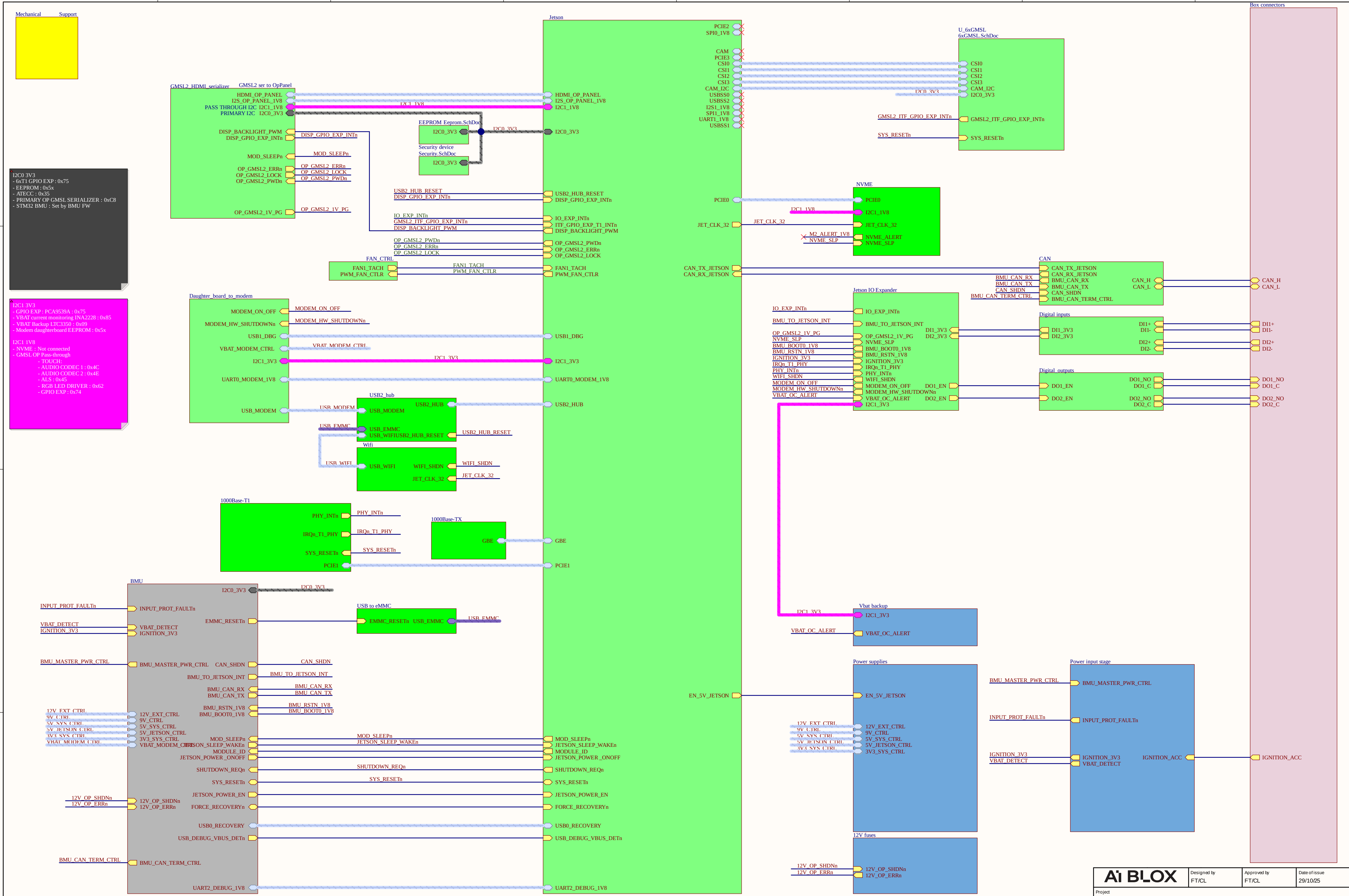

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